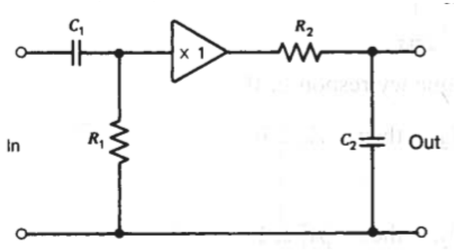
1 Pre-Lab Conceptual Questions

1. Describe at least two types of ADCs. Explain how they work.
2. What is SCA and what is MCA? How do they work?
3. Derive the output response of the CR-RC circuit shown below to a step voltage pulse of amplitude V_0 at t = 0. What happens if $R_1C_1 = R_2C_2$?



4. Explain time jitter and amplitude walk. What are they? How are they different?

2 The Single-Channel Analyzer

2.1 Equipment Setup

2.1.1 Required Equipment

- oscilloscope
- pulse generator
- amplifier
- single-channel analyzer (SCA)
- MCA

2.1.2 Measurement Setup

Turn on the NIM Bin

- Connect the output of the pulser to the input of the amplifier
- Split the amplifier output. Connect to the SCA input and to the channel 1 input on the scope.
- Connect the SCA output to the channel 2 input on the scope.
- PULSER: Set the amplitude to ~0.1 V (or as low as possible), positive polarity, fast leading edge, 200 µs tail
- AMPLIFIER: Choose unipolar pulses and adjust the gain to produce pulses of positive amplitude between 0 and 10 V
- SCA: Set to integral mode so that it acts as a simple threshold discriminator
- SCOPE: Set the trigger threshold level low enough so that the output logic pulse is displayed

2.2 SCA Measurements

2.2.1 Lower-Level Discriminator Measurements

- Measure rise time of the pulse.
- GOAL: visually inspect the range of input pulses that cause the SCA to generate an output logic pulse
- Vary the lower level dial to determine the lower-level discriminator (LLD) setting at which the input pulse amplitude is discriminated
- For a range of pulser amplitudes determine the maximum LLD setting that produces logic pulses from the SCA. For each amplitude, measure the rise time of the signal going in to the SCA.
- Be sure to investigate the ends of travel of the LLD potentiometer (What is the max voltage that can be discriminated? What is the min voltage that can be discriminated?).
- Plot the SCA input amplitude vs. the maximum LLD setting.

2.2.2 SCA Window Mode Measurements

- Switch the SCA to *window* mode
- NOTE: For window mode, an upper level setting of 10 corresponds to a 1.0V window width
- Select a reasonable LLD setting